

OOP



Project

ENG. Nadeen Anwar

Project Title:

CareSync ✨

(Smart Medication Reminder System)

(نظام رعاية صحية ذكي)

Problem Definition:

Managing medication schedules can be challenging, especially for elderly patients and individuals with chronic conditions. Many people rely on memory or handwritten notes, which often leads to missed doses or incorrect timing.

This issue, known as medication non-adherence, can negatively affect health and increase stress for patients. Therefore, there is a clear need for a reliable and easy-to-use system that helps users organize their medications and receive timely reminders

Recommended Solution:

CareSync is a desktop application developed using C# and Windows Forms, designed to simplify and organize medication management.

The system is built using Object-Oriented Programming (OOP) principles to ensure a clean structure and ease of maintenance.

Key features include:

- *Medication Management*: Support for adding, viewing, updating, and deleting medications with their dosage and schedule.
 - *Notification System*: Audio and visual alerts that notify users at the correct time based on the system clock.
 - *Digital Archive*: Store prescription images and doctor contact details for quick access.
 - *Persistent Storage*: Data is saved using file handling to ensure it remains available across sessions.
 - *User-Friendly Interface*: A simple and clear design suitable for users of all experience levels.
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Objectives:

1. *Improve Medication Adherence*: Help users take their medication on time with fewer mistakes.
 2. *Organize Medical Information*: Provide a centralized place to manage medications and prescriptions.
 3. *Ensure Data Reliability*: Use structured programming techniques to safely store and retrieve data.
 4. *Reduce User Stress*: Make medication management easier and more comfortable for users.
 5. *Demonstrate Technical Skills*: Apply OOP, file handling, and event-driven programming in a practical project.
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Team Members

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